

Tenzin Kunsang

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PROFESSIONAL EXPERIENCE

Machine Learning Engineer, *Transmural Systems*

Jul 2024 – Dec 2024 | Andover, MA

- Developed robust multi-algorithm detection pipeline to solve critical stability issues in the company's cardiac device system, replacing inconsistent YOLO baseline and achieving 87% accuracy with sub-100ms response times for MRI-guided procedures
- Designed and benchmarked CV algorithms (Frangi filters, Kalman tracking, KAZE features) against YOLO variants to overcome production deployment blockers, improving IoU from 0.58 to 0.83 while meeting clinical performance standards
- Built detection and tracking models through 8 iterative cycles with MRI and CV specialists to ensure clinical viability, refining algorithms based on real-world performance requirements that prevented hospital deployment
- Designed integrated monitoring platform with multi-target tracking interface to enable device commercialization, providing visualization capabilities required to transition from research prototype to marketable medical device

Lead Data Engineer / Software Developer, *ICF*

Jul 2020 – Jul 2023 | Reston, VA

- Led pioneering data integration initiative to solve fragmented veteran employment tracking across 3 federal systems, building the first unified dataset enabling complete 200K+ service member journey analysis previously impossible with scattered data
- Created Tableau dashboards mapping veteran career transitions with skills-matching visualizations, geographic placement patterns, and time-to-employment funnels to empower partners to identify successful pathways and optimize their support services
- Developed predictive analytics platform to demonstrate program impact and secure continued funding, building Tableau models showing 63-day faster employment outcomes and 11% higher wages that justified the \$2.74M program investment
- Engineered scalable SQL data models to overcome legacy system limitations preventing comprehensive reporting, enabling efficient analysis across 35+ installations and delivering \$50K in annual operational savings through automated processes
- Built data engineering capabilities from zero by establishing team's first data-focused infrastructure, mentoring 3 junior developers while creating repeatable workflows that transformed ad-hoc reporting into systematic data operations

Graduate Student Teaching Assistant, *Northeastern University*

Jan 2024 – Apr 2025 | Boston, MA

- *Pattern Recognition & Computer Vision (Spring '25)* - Provided mentorship on OpenCV and C++ and technical concepts including image feature extraction, morphological operations, and AR/VR to students with varying levels of background knowledge
- *Algorithms & Data Structures (Spring and Summer '24)* - Provided mentorship on Git, GitHub, CoPilot, Python, C, and concepts including dynamic programming, graph and search algorithms, and amortized analysis

SKILLS

ML Libraries & Tools: PyTorch, TensorFlow, OpenCV, Scikit-learn, Pandas, SciPy, Hugging Face Transformers, Attention Mechanisms, Multi-modal Learning, Parameter-Efficient Fine-tuning (PEFT), RAG, Vector databases, Visualizations (Matplotlib, Seaborn, OpenCV)

Certifications: Neural Networks and Deep Learning , Machine Learning with Python , Structuring Machine Learning Projects , IBM Data Science Professional Certificate 

Development: Python, Java, C++, JavaScript, SQL, R, Git, CI/CD, Jenkins, Source Control Management

PROJECTS

Referring Image Segmentation, *Parameter-Efficient Finetuning*

Apr 2025

Designed a multimodal RIS model (image and text inputs) that uses Dense Adapters with state space models and Transformer architecture while updating only 5.8% of the backbone parameters. "Most Innovative Project" award in 54-student graduate class.

Depth Estimation for Indoor Scenes, *Monocular DE*

Dec 2024

- Developed a lightweight semi-supervised adaptation of the Depth Anything framework using a teacher-student approach with data augmentations for indoor scenes.

Plant trait classification, *Deep CNN networks, ensemble, & multi-modal models*

Apr 2024

- Multi-modal deep learning system combining ResNet50, InceptionV3, and EfficientNet to classify plant traits using text, numerical, and image data streams.

Spacecraft Telemetry Anomaly Detection, *Time Series Deep Learning*

Feb 2024

- Developed Convolutional Autoencoders and Temporal Convolutional Networks for anomaly detection in spacecraft telemetry data, processing 1.8M+ data points with strong TCN performance (MSE: 0.013-0.191)

EDUCATION

Northeastern University,

Sep 2023 – May 2025 | Boston, MA

Master of Science in Computer Science, Specialization in AI & Data Science

Relevant Coursework: Advanced Perception, Pattern Recognition & Computer Vision, Algorithms, Deep Learning, Machine Learning, Unsupervised Machine Learning, Mobile Application Development, Program Design Paradigm

Carleton College, Bachelor of Arts in Mathematics

Sep 2016 – Jun 2020 | Northfield, MN

Relevant Coursework: Artificial Intelligence, Real-Time Systems, Data Structures, Probability, Applied Regression Analysis, Statistical Consulting, Data Science, Linear Algebra